

TPIAMS

THEFT PREVENTION INTELLIGENT ALARM MONITORING SYSTEM

Real-Time Safety, Anytime, Anywhere

An intelligent, industrial-grade security platform engineered to protect telecom tower assets from theft, intrusion, and infrastructure loss — with 24×7 SOC-backed remote monitoring, AI-enhanced alarm intelligence, and multi-channel alerting.

- TPIAMS V7.0
- INDUSTRIAL GRADE
- SOC ENABLED
- AI MONITORING

PROTECTION COVERAGE

- Battery Theft Prevention
- Cable & Copper Theft
- DG / Diesel Theft
- Radio Equipment Theft
- Intrusion Detection
- Solar Panel Protection
- Infrastructure Theft
- Fire & Tamper Detection

13+

Years Experience

300+

Field Engineers

24/7

SOC Monitoring

PAN

India Operations

01 PRODUCT OVERVIEW

TPIAMS (Theft Prevention Intelligent Alarm Monitoring System) is Shroti Telecom's flagship next-generation security solution purpose-built for telecom tower sites, remote infrastructure, and industrial deployments. It integrates a rugged hardware controller with cloud-connected software and AI-enhanced alarm intelligence, backed by a dedicated 24×7 Security Operations Center (SOC), to deliver end-to-end asset protection. The system monitors for battery theft, diesel fuel pilferage, cable tampering, equipment removal, and unauthorized intrusion in real time — triggering multi-channel alerts instantly upon detection. Engineered for the harshest field conditions, TPIAMS operates autonomously with internal battery backup and requires minimal on-site intervention.

WHAT TPIAMS PROTECTS:

- Battery Theft Prevention
- Cable & Copper Theft
- DG / Diesel Theft
- Radio Equipment Theft
- Intrusion Detection
- Solar Panel Protection
- Infrastructure Thft
- Fire & Tamper Detection

02 KEY FEATURES

REAL-TIME MONITORING

Continuous site surveillance with instant event detection and automated response workflows.

AI ALARM INTELLIGENCE

Smart alarm filtering with pattern recognition reduces false alarms and prioritizes genuine threats.

16 ALARM INPUTS

Multi-sensor support: door, motion, fire, tamper, battery & power inputs on a single controller.

MODBUS RS485

Industry-standard serial communication for reliable, noise-resistant long-distance integration.

MULTI-CHANNEL ALERTS

WhatsApp, SMS & Robo Voice Call notifications dispatched instantly on alarm events.

BUILT-IN HOOTER

125 dB high-power on-site alarm for active theft deterrence and intruder warning.

BATTERY BACKUP

1200 mAh internal battery with BMS — up to 10 hours autonomous backup operation.

CLOUD & API READY

Seamless cloud integration with REST API support for enterprise and IoT platforms.

HARSH ENVIRONMENT

Rated -10°C to +55°C. Rugged design for outdoor & remote telecom deployments.

REMOTE DASHBOARD

Web and mobile dashboard for live site monitoring, analytics and alarm history.

SOC INTEGRATION

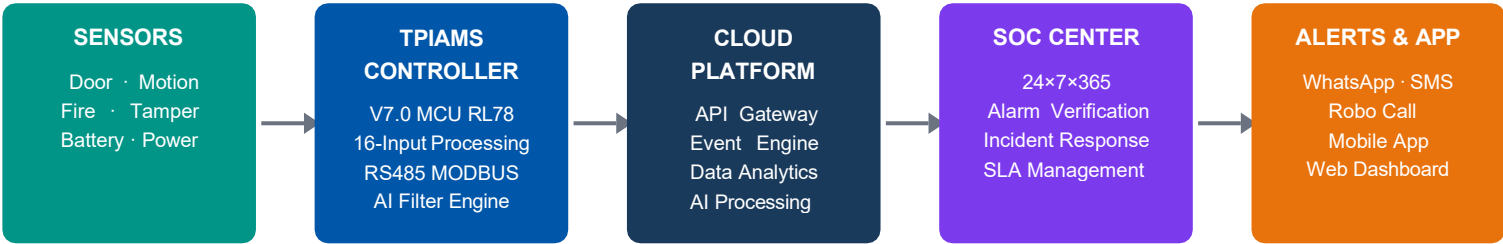
24x7x365 Security Operations Center with SLA-driven incident response and escalation.

LIGHTNING PROTECTION

Built-in lightning surge protection on infrared sensor inputs for exposed installations.

03

SYSTEM ARCHITECTURE



Field-deployed sensors feed alarm data into the TPIAMS V7.0 controller via wired inputs. The controller processes events locally using its AI filter engine — suppressing false alarms, triggering the 125 dB hooter — and simultaneously transmits structured data to the Shroti Cloud over cellular/RS485. The cloud platform applies AI-based event verification and routes confirmed alarms to the SOC Command Centre, which dispatches multi-channel alerts (WhatsApp, SMS, Robo Call, App notification) and escalates field response

04

TECHNICAL SPECIFICATIONS

SYSTEM OVERVIEW	
Processor	16-bit MCU – Renesas RL78
Input Voltage	48V DC
Operating Temperature	–10°C to +55°C
PHYSICAL SPECIFICATIONS	
Weight	1.57 kg
Antenna	Internal & External support
POWER & BACKUP	
Backup Duration	Up to 10 hours (under typical load)
ALARM & MONITORING	
Supported Sensors	Door, Motion, Fire, Tamper, Battery, Power monitoring
Local Alarm	Built-in Hooter — 125 dB high-power
AI Filter Engine	Intelligent alarm pattern recognition & false-alarm suppression
COMMUNICATION	
Alert Channels	Robo Call, WhatsApp, SMS, Mobile App push notifications
Cloud Integration	REST API support; cloud-hosted dashboard & analytics
INDICATORS	

05 APPLICATIONS

Telecom BTS & Tower Sites

Battery, DG, cable, and RRU equipment protection for 2G/4G/5G sites. AI-based

Battery Swapping Stations

Real-time monitoring of high-value Li-ion and VRLA battery banks with instant theft

Solar Energy Sites

Dual-logic solar panel protection using NO/NC resistance wire sensing with

Remote Infrastructure

Unmanned shelter, outdoor enclosure & cabinet security in remote or off-grid areas.

Industrial Security

Perimeter protection, fire detection & tamper monitoring for industrial assets and

IoT & Smart Sites

Integration into broader IoT automation frameworks via REST API and MODBUS

06 BUSINESS BENEFITS

Theft Reduction

Deter and detect asset theft before losses occur with

Cost Savings

Significant reduction in replacement & downtime

Remote Visibility

Full site awareness from any location via web

Faster Response

Immediate multi-channel alerts minimize response

24x7 SOC

Round-the-clock monitoring by trained security analysts

07 SECURITY OPERATIONS CENTER (SOC)

SOC CAPABILITIES

- 24x7x365 centralised monitoring of all field sites
- Real-time detection: theft, fire, and communication loss
- AI-assisted alarm verification reduces false dispatches
- Automated WhatsApp, SMS & Voice Call notifications
- Escalation to field teams within defined SLA timelines
- Incident verification, ticketing, and closure tracking
- Proactive monitoring using historical telemetry trends
- Remote dashboard access with admin-level control

24/7

CONTINUOUS
MONITORING

365

DAYS PER YEAR

Instant

ALERT
ACKNOWLEDGEMENT

SLA

ESCALATION
PROTOCOL

08 ALERTS & NOTIFICATIONS

WhatsApp Alerts

Instant message notification to registered contacts with full event details, site ID, and timestamps.

SMS Alerts

Carrier-independent SMS dispatch ensuring delivery even in low-data or low-signal environments.

Robo Voice Call

Automated voice calls to on-call personnel for high-priority alarm escalation and acknowledgement.

Mobile App Push

Real-time push notifications and live event feed via the Shroti Telecom TPIAMS admin application.

09 COMPLIANCE & RELIABILITY

Industrial-Grade Design

Built to withstand demanding telecom field environments with rugged enclosure and robust internal components rated for continuous operation.

Weather-Resistant

Designed for outdoor and semi-outdoor deployment, tolerating temperature extremes (-10°C to +55°C), dust, and high humidity conditions.

Lightning Protection

Infrared sensors feature built-in lightning surge protection for installations in exposed outdoor locations near telecom towers.

Stable Communication

RS485 MODBUS protocol ensures long-distance, noise-resistant data transmission in electrically harsh environments with interference.

Battery Management System

Intelligent BMS protects the internal backup battery from over-charge, over-discharge, and thermal runaway events continuously.

Standalone Operation

TPIAMS operates as a fully autonomous unit with internal AI processing; no external controller required for core protection functions.

10 COMPANY PROFILE

SHROTITELECOM

Established 2012 · Bhopal, M.P.

13+ Years in Telecom

300+ Field Engineers

24/7 SOC Centre

PAN India Ops

Established in 2012, Shroti Telecom began as a passive infrastructure solutions provider with a vision to deliver rugged, easy-to-integrate, and cost-effective products for telecom tower operations. Over 13+ years, it has transformed into a full-stack telecom infrastructure and security solutions company with several first-in-India innovations including TPIAMS.

Theft Prevention (TPIAMS)

Remote Monitoring (RMS)

IoT Automation

Telecom Shelter Mfg.

Smart Power Electronics

Field O&M Services

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